

Research Article

Track My Device With AI Assistant System for Real-Time Location Monitoring

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Abstract— The loss or theft of mobile devices such as smartphones, tablets, and laptops can lead to serious consequences including data exposure, financial loss, and privacy violation. Existing tracking solutions often provide only basic location services with limited predictive capability and fragmented functionality. This research aims to design and develop a Track My Device with AI Assistant system that integrates real-time GPS tracking, machine learning-based movement prediction, geofencing alerts, SIM change notifications, low battery warnings, remote lock, alarm activation, and an AI chatbot into a single unified platform. The system uses GPS technology for continuous location monitoring and machine learning algorithms to analyze movement history and predict possible future device locations. An intelligent alert system notifies users immediately when suspicious events such as SIM card replacement, device shutdown, or unauthorized movement occur. The AI chatbot enables users to interact with the system through natural language queries to request location updates, device status, or activate security features. A multi-device management dashboard allows users to monitor all their devices from one interface displaying location, battery level, and security status. The proposed system is expected to improve device recovery rates, strengthen anti-theft protection, and provide users with a convenient and intelligent solution for managing multiple devices in real time.

Keywords— Device Tracking, Artificial Intelligence, GPS Technology, Machine Learning, Geofencing, Anti-Theft Systems, Remote Locking, AI Chatbot, Multi-Device Management

Graphical Abstract-



1. Introduction

The growing dependence on mobile devices including smartphones, tablets, and laptops has made device security, tracking, and recovery a critical concern in modern digital environments. These gadgets hold personal, financial, and professional data, and their loss can lead to privacy violations, financial harm, and disruption of daily routines. When a device goes missing or is stolen, the owner may face not only the expense of replacement but also the danger of losing sensitive information and access to vital services. As digital device usage continues to rise, so too does the likelihood of misplacement, theft, and unauthorized entry[1].

This issue is especially serious given that today's devices store extensive personal, academic, and professional records, which could be misused if accessed by unauthorized individuals. In many instances, conventional tracking approaches fail to deliver prompt notifications or precise location updates, particularly when network connectivity is weak or when the device has been deliberately powered off, thereby diminishing recovery prospects and heightening security vulnerabilities[2].

Although numerous tracking and security applications exist, many offer restricted functionality, limited predictive power, and inadequate response to critical events like SIM card changes, power offs, low battery situations, and unauthorized movement. Current systems often function as isolated solutions that handle only specific tasks such as basic location tracking or remote locking, without delivering a complete framework that combines real time tracking, predictive analytics, smart alerts, and conversational interfaces[3]. Recent investigations have examined the application of artificial intelligence and machine learning to device tracking, showing that intelligent models can enhance the ability to

recover lost or stolen items by detecting location patterns and unusual movements. GPS enabled real-time monitoring platforms have also been created to consistently capture location information and relay it to a centralized interface, yet many of these systems rely heavily on stable network connections and may underperform in environments with weak signals[4]. Studies have further revealed that automated alert systems incorporating geofencing, SIM card change detection, and battery status warnings can greatly improve device protection, particularly when such alerts are provided instantly and paired with straightforward response mechanisms. Machine learning methods have been explored for analyzing movement histories and forecasting future location trends based on user habits, while conversational AI and chatbots have been shown to make digital platforms more intuitive, responsive, and accessible to non-technical users. Nevertheless, despite these developments, the literature indicates that very few systems offer a comprehensive framework that merges real-time tracking, predictive analysis, smart alerts, chatbot communication, and multi-device oversight into a single platform, leaving users with fragmented tools that fail to deliver complete protection and efficient recovery assistance[5].

This study is important because it addresses the growing need for an intelligent, integrated, and user-friendly solution that can effectively mitigate the risks associated with device loss and theft, particularly in regions where users may not have access to advanced security technologies. By combining artificial intelligence, real-time monitoring, and conversational interfaces, this research contributes to the advancement of mobile security systems and provides practical insights for developers, policymakers, and end-users seeking reliable and accessible device protection strategies. To bridge these gaps, this research aims to create a Track My Device with AI Assistant system that unifies real-time GPS tracking, machine learning-driven movement forecasting, geofencing notifications, SIM change alerts, low battery warnings, remote locking, alarm activation, and an AI-powered chatbot within a single cohesive platform. The specific objectives of this study are: to build a real-time device tracking system using GPS technology that ensures dependable location monitoring even in low-connectivity areas; to create an AI-powered monitoring system that examines device movement history and applies machine learning techniques to estimate the likely future position of the device; to establish an intelligent alert and security framework that notifies users when the device exits a designated geographic zone, when the SIM card is swapped, when the device is switched off, or when the battery becomes critically low, while also incorporating anti-theft functions such as remote lock and alarm, and integrating an AI chatbot for conversational engagement; and to build a user-friendly multi-device management interface that enables users to oversee multiple devices through an interactive dashboard showing location, status, and battery information.

1.1 Objectives of The Study

1.1.1 Main Objective

To design and develop an AI-powered smart device tracking system that enables real-time location monitoring, intelligent

movement prediction, automated security alerts, and multi-device management through a unified platform with conversational AI assistance.

1.1.2 Specific Objectives

1. To develop a real-time device tracking system using GPS technology that ensures accurate location monitoring even in areas with weak network connectivity.
2. To design an AI-powered monitoring system that analyzes device movement history and uses machine learning techniques to predict the possible future location of the device.
3. To implement an intelligent alert and security system that notifies users when the device leaves a defined geographical area, when the SIM card is changed, when the device is turned off, or when the battery level is low, while also providing anti-theft features such as remote lock and alarm, and integrating an AI chatbot for conversational interaction.
4. To create a user-friendly multi-device management system that allows users to monitor multiple devices through an interactive dashboard displaying location, status, and battery information.

1.2 Organization of The Article

This paper is organized into five main sections. Section 1 provides the introduction, which outlines the research background, problem statement, objectives, and the overall structure of the paper. Section 2 presents a comprehensive review of related literature on device tracking technologies, including AI-powered tracking systems, GPS-based monitoring, intelligent alert mechanisms, machine learning for movement prediction, AI chatbots, and multi-device management dashboards. Section 3 describes the theoretical foundations and calculation methods used in the development of the system, covering GPS distance calculation, Kalman filter for movement prediction, geofencing alert logic, evaluation metrics, and loss function for the movement prediction model. Section 4 presents the experimental method, including the study design, system architecture, flowchart, dataset collection, model architecture and training, testing procedures, hardware and software environment, evaluation metrics, and ethical considerations. Section 5 presents the results and discussion of the study, covering system performance metrics such as GPS tracking accuracy, movement prediction error, geofencing detection rates, AI chatbot performance, user acceptance survey findings, and multi-device dashboard functionality, followed by an interpretation of the findings in relation to existing literature, strengths, limitations, implications for practice and policy, and future research directions. Section 6 concludes the manuscript with a summary of the key findings, acknowledges the limitations of the study, and provides recommendations for future research and system improvement.

2. Related Work

2.1 AI-Powered Device Tracking with LLMs

Recent progress in large language models (LLMs), such as OpenAI's ChatGPT and Google's Bard, has significantly

reshaped how users interact with digital systems. These models have opened up new possibilities for combining intelligent conversational agents with wireless sensor networks (WSNs), enabling more adaptive and context-aware tracking solutions. A representative example is ChatTracer, an LLM-powered system designed for real-time Bluetooth device tracking. The system comprises three main components: a distributed array of Bluetooth sniffing nodes that capture device signals, a centralized database for storing and organizing captured data, and a fine-tuned LLM that interprets the signals and provides meaningful responses to users[6]. The development of ChatTracer was motivated by the observation that both Apple and Android devices continuously broadcast hundreds of Bluetooth Low Energy (BLE) packets per minute, even in idle states, making them traceable without requiring active user engagement. The system introduces two major innovations: an efficient algorithm for grouping BLE packets, and a fine-tuning strategy for the LLM that integrates supervised fine-tuning (SFT) with reinforcement learning from human feedback (RLHF), resulting in improved accuracy and contextual relevance[7]. Experimental results show that ChatTracer not only outperforms conventional localization techniques but also offers a more intuitive user experience by allowing natural language interaction. In addition, LLMTrack has been proposed to demonstrate how LLMs can perform zero-shot trajectory recognition using a single-prompt methodology that combines role-play and step-by-step reasoning with raw inertial measurement unit (IMU) data. Furthermore, Wi-Chat has been introduced as the first LLM-powered Wi-Fi-based human activity recognition system, capable of interpreting raw Wi-Fi signals by embedding sensing principles directly into prompts, thus bypassing traditional signal processing steps[2]. These studies collectively illustrate that integrating LLMs with wireless sensing technologies holds considerable promise for advancing the capabilities of modern tracking and monitoring systems, making them more intelligent, responsive, and user friendly.

2.2 GPS and GSM-Based Real-Time Tracking Systems

Satellite-based positioning technologies, commonly referred to as GPS, alongside mobile communication standards such as GSM, have become essential tools for tracking the real-time movements of vehicles and portable devices across transportation, logistics, and personal security sectors[8]. These tracking solutions operate by capturing satellite signals to calculate the exact location of a device, after which the coordinates are relayed through cellular networks to a central platform or directly to the user. Research has shown that GPS and GSM based tracking solutions effectively support users in monitoring their belongings using existing telecommunication infrastructures, offering both affordability and reliability[9]. A typical system architecture includes GPS receivers that acquire location data, including latitude and longitude, and transmit this information to the user's device upon request via SMS or internet-based communication. Studies have evaluated the performance of GPS sensors in diverse settings, ranging from densely built urban centers to open rural landscapes, to understand how environmental and infrastructural factors influence positioning accuracy[10]. In

certain cases, cloud-based tools such as Google Sheets have been used as storage repositories for GPS logs, where data is sent over GPRS networks to enable real-time tracking and remote data access. Once received, the data is displayed on interactive maps, giving users a clear view of their device's current position. More sophisticated methods combine GPS readings with Wi-Fi and Bluetooth signals to improve device localization accuracy. These systems assess whether a device meets predefined conditions based on location probability, movement speed, and wireless signal patterns, thereby identifying whether the device should be treated as a primary entity within a monitored zone[11]. The integration of GPS, GSM, and complementary wireless technologies has thus laid a strong foundation for modern tracking systems, offering flexibility, adaptability, and enhanced precision in a variety of real-world scenarios.

2.3 Intelligent Anti-Theft and Alert Systems

Intelligent anti-theft mechanisms have undergone significant advancements in response to the increasing threats of device theft and unauthorized access. Modern systems now integrate multiple sensing technologies, including vibration detectors and infrared sensors, to provide more robust and reliable protection[12]. These systems are typically designed to operate in a multi-stage defense mode, where an initial trigger such as sudden movement or proximity detection activates a warning signal, while persistent tampering leads to full alarms and automatic notifications to the device owner via mobile networks[13]. By combining different types of sensors in a logical configuration, false alarms are significantly reduced, as alerts are only generated when multiple sensors are triggered simultaneously. In addition, predictive theft notification systems have been developed for retail and personal environments, leveraging imaging devices and biometric matching to anticipate potential theft incidents before they occur. These systems perform real-time facial recognition and cross-reference against databases of known offenders, enabling proactive security responses rather than reactive measures[14]. The incorporation of such intelligent features into anti-theft systems not only improves detection accuracy but also enhances user confidence and reduces response time during security breaches, making them increasingly valuable for both individual and commercial applications.

2.4 Machine Learning for Movement and Trajectory Prediction

Machine learning techniques have become essential tools for modeling and predicting the movement patterns and future trajectories of moving objects across various domains. The rapid expansion of positioning technologies and GPS-equipped devices has generated enormous volumes of trajectory data, which serve as a valuable resource for analytical processes such as short-term forecasting of future positions and behavior modeling[15]. A comprehensive review of more than 50 studies on predictive analytics for moving objects offers a detailed classification of prediction algorithms and highlights the key challenges that emerge when transitioning from conventional data processing methods to Big Data environments. This review encompasses

a wide range of moving entities, including pedestrians, vehicles, aircraft, marine vessels, and wildlife, and examines both offline clustering approaches and online real-time prediction techniques. Another significant contribution introduced a technique for forecasting the motion paths of movable objects using adversarial neural network models[16]. This method involves the collection of environmental and object data, which are then fed into an adversarial neural network architecture. The model computes performance scores and iteratively adjusts its parameters until it meets the required accuracy thresholds, enabling fast and precise trajectory prediction even in dynamic and complex settings. Furthermore, deep learning based object recognition and identification models have been developed to improve tracking accuracy by performing identity detection and verification through similarity comparisons of object identifiers[2]. These advancements collectively demonstrate the growing potential of machine learning in enhancing the reliability and efficiency of movement prediction systems, with implications for traffic management, surveillance, autonomous navigation, and personal device tracking.

2.5 AI Chatbots and Conversational User Interfaces

Conversational AI has emerged as a transformative technology for improving user interaction with tracking and security systems. LLM powered chatbots enable users to interact with complex tracking systems through natural language rather than navigating technical menus. ChatTracer demonstrates this by providing an intelligent interface for user interaction where users can query device status, request location updates, and trigger security actions through conversational commands[17]. The use of supervised fine-tuning combined with reinforcement learning from human feedback during model training has been found to significantly enhance the accuracy and relevance of responses, especially for tracking and security-related inquiries. Research has also demonstrated that conversational interfaces can make digital systems more intuitive and responsive for non-technical users, particularly during emergencies when quick access to information is critical[18]. These systems are designed to understand context, maintain conversation history, and provide appropriate responses based on device status and user permissions[19]. The integration of chatbots into tracking applications not only simplifies user interaction but also reduces the learning curve associated with complex security tools, making them more accessible to a wider audience. Furthermore, the ability to support multiple languages, including Swahili, extends the reach of these systems to diverse user populations, addressing language barriers that often hinder technology adoption in regions such as East Africa. As conversational AI continues to evolve, its role in enhancing user experience and operational efficiency in device tracking systems is expected to grow significantly.

2.6 Multi-Device Management Dashboards

Centralized dashboard interfaces for managing multiple connected devices have become increasingly important as users now own a growing number of smartphones, tablets, laptops, and other smart devices. These dashboards provide a unified interface that allows users to monitor device location,

battery status, security alerts, and connectivity from a single point of access[20]. Dashboard systems typically offer features such as widget customization, layout personalization, PDF export, email views, and real-time data updates, enabling users to effectively track the health and status of their devices and servers. For efficient multi device management, each device must first be registered and integrated into the system before it can be monitored through dashboard gadgets[21]. Research indicates that dashboard-based management significantly improves monitoring efficiency, especially for users who need to oversee multiple devices simultaneously. However, many existing platforms lack sufficient integration between status monitoring, alert generation, and control functions, resulting in fragmented user experiences. The proposed Track My Device with AI Assistant system addresses these shortcomings by offering a unified dashboard that consolidates real-time location tracking, battery levels, security alerts, and detailed device status information[22]. This integration not only enhances user convenience but also enables faster decision-making and more effective responses to security incidents. Furthermore, the dashboard is designed to be user-friendly, with intuitive navigation and visual indicators that help users quickly identify devices requiring attention. By providing a comprehensive overview of all registered devices, the system empowers users to maintain greater control and awareness over their digital assets, which is particularly valuable for families, small businesses, and organizations managing multiple devices.

2.7 Security and Privacy in Device Tracking

Device tracking systems collect and process a wide range of sensitive user information, including real-time geographic location, device identifiers, movement patterns, and behavioral trends. While this data is essential for enabling effective tracking and recovery functionalities, it also presents significant security and privacy risks if not properly protected [28]. Unauthorized access to tracking data can lead to serious consequences such as stalking, identity theft, corporate espionage, and surveillance without user consent. Several studies have highlighted the vulnerabilities associated with location-based services, including insecure data transmission, weak authentication mechanisms, and insufficient access control policies [29]. In many cases, attackers can intercept location data transmitted over unsecured networks or exploit vulnerabilities in mobile applications to gain unauthorized access to user information. Encryption protocols such as Transport Layer Security (TLS) and end-to-end encryption are widely recommended to secure data during transmission, while secure storage practices are essential to protect location histories from being accessed by malicious actors or unauthorized insiders [30]. Additionally, user awareness and consent mechanisms are critical in ensuring transparent handling of personal data, especially in multi-device environments where tracking may be performed across shared accounts or family groups. Despite the availability of these protective measures, research indicates that many commercial tracking applications continue to fall short of industry security standards, leaving users exposed to both external cyber threats and internal misuse [31]. To address these

challenges, privacy-preserving techniques such as differential privacy, data anonymization, and federated learning are being explored to allow useful data analysis without compromising individual user privacy. These approaches aim to balance the operational benefits of device tracking with the fundamental rights of users to control their personal information, thereby fostering greater trust and wider acceptance of tracking technologies in everyday life.

2.8 User Acceptance and Usability of Tracking Systems

The successful adoption of device tracking systems depends not only on their technical performance but also on how users perceive their usefulness, ease of use, and overall reliability [32]. Numerous studies have demonstrated that users are more inclined to adopt and consistently use tracking solutions when they perceive them as intuitive, dependable, and beneficial to their daily activities, whether for personal security, device recovery, or family monitoring [33]. Key factors influencing user acceptance include the complexity of the system interface, the level of trust in data security and privacy practices, user awareness of available features, and the perceived value provided by the service. In developing regions, additional barriers such as limited digital literacy, language constraints, and concerns about data misuse can significantly hinder adoption rates [34]. Therefore, designing simplified, user-friendly interfaces, providing clear and accessible privacy policies, and offering multilingual support are essential strategies for improving user engagement and long-term satisfaction. Furthermore, interactive feedback mechanisms, onboarding tutorials, and ongoing user training have been identified as important contributors to sustaining system usage and fostering positive user experiences [35]. Understanding these usability and acceptance factors is critical for developers, service providers, and policymakers aiming to promote the widespread and equitable adoption of tracking technologies across diverse user populations and geographic contexts. By prioritizing user-centered design and addressing usability challenges, tracking systems can become more inclusive and effective tools for device protection and security.

3. Theory and Calculation

The Track My Device with AI Assistant system relies on several mathematical and algorithmic foundations to enable real-time GPS tracking, movement prediction, geofencing alerts, and performance evaluation. This section presents the theoretical basis and calculation methods used in the development of the system.

3.1 GPS Distance Calculation (Haversine Formula)

The system uses GPS technology to continuously monitor the location of mobile devices. To determine whether a device has moved outside a defined safe area or for computing the distance between two geographic positions, the system applies the Haversine method, which determines the shortest path between two points on a sphere using their latitude and longitude values, is expressed as:

$$c = 2 \times a \tan 2 \left(\sqrt{a}, \sqrt{(1-a)} \right)$$

$$d = R \times c$$

Where φ_1 and φ_2 are the latitudes of the two points in radians, $\Delta\varphi$ is the difference between the two latitudes, ∇y is the difference between the two longitudes, R is the Earth's mean radius (approximately 6,371 kilometers), and d is the calculated distance between the two points. This calculation is essential for determining how far a device has moved from its last known position and for triggering geofence alerts when a device exceeds a predefined distance threshold from a safe zone[11].

3.2 Movement Prediction with Kalman Filter

To predict the future location of a lost or stolen device, the system employs a Kalman filter algorithm. The Kalman filter is a recursive mathematical algorithm that estimates the state of a moving object based on a series of incomplete and noisy measurements. For device tracking, the system uses historical GPS data including position, velocity, and acceleration to predict where the device is likely to move next. The Kalman filter operates in two main steps: prediction and update. In the prediction step, the algorithm estimates the current state based on the previous state using the equation:

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$

Where $\hat{x}_k$ is the predicted state estimate, A is the state transition matrix, $\hat{x}_{k-1}$ is the previous state estimate, B is the control input matrix, and u_k is the control vector. The second step is the update step, where the algorithm corrects the predicted estimate using the actual measurement:

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$$

Where K_k is the Kalman gain, z_k is the actual measurement, and H is the measurement matrix. This prediction capability enables the system to provide users with an estimated future location of their device even when real-time GPS updates are temporarily unavailable due to network or connectivity issues. The algorithm is particularly effective for tracking objects moving along predictable paths based on user routines and historical movement patterns[23].

3.3 Geofencing Alert Logic

Geofencing is a core anti-theft feature that creates virtual boundaries around defined safe areas such as home, workplace, or school. When a device enters or leaves these virtual perimeters, the system automatically triggers alerts. The mathematical logic behind geofencing involves calculating whether a device's current GPS coordinates lie inside or outside a defined circular geofence. The system determines the distance between the device's current location

and the center point of the geofence using the Haversine formula described in Section 3.1. If this distance is less than or equal to the geofence radius, the device is considered inside the safe area. If the distance exceeds the radius, the device has left the geofence and an alert is triggered. For predictive geofencing, the system also calculates the time remaining before a device is expected to cross the geofence boundary based on its current speed and direction. The time-horizon calculation is expressed as:

$$t = \left(\frac{d_o - r}{v} \right)$$

Where d_o is the current distance from the geofence center, r is the geofence radius, and v is the device's current velocity. This predictive capability allows the system to notify users before a device actually leaves the safe area, enabling proactive security measures. The system also supports geofencing without GPS using pedometer-based algorithms that estimate location changes based on step counting and movement patterns[11].

3.4 Evaluation Metrics for Tracking Performance

To assess the performance of the Track My Device with AI Assistant system, four standard evaluation metrics are used. Accuracy measures the overall correctness of the tracking system in correctly identifying device locations and triggering appropriate alerts. It is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{(TP + TN + FP + FN)} \times 100\%$$

Precision measures how reliable the system is when it predicts a device location or triggers an alert. A high precision means when the system reports a device location or threat, it is usually correct. It is calculated as:

$$\text{Precision} = \frac{TP}{(TP + FP)} \times 100\%$$

Recall measures how well the system catches actual device movements or security events. A high recall means the system rarely misses a device location update or security threat. It is calculated as:

$$\text{Precision} = \frac{TP}{(TP + FP)} \times 100\%$$

The F1-Score offers a balanced measure that combines both precision and recall, making it especially valuable when assessing tracking systems where both false positives and false negatives have significant consequences. It is calculated as:

$$\text{F1 - Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Where TP (True Positive) represents correctly identified device locations or correctly triggered alerts, TN (True Negative) represents correctly identified safe conditions, FP (False Positive) represents incorrectly triggered alerts, and FN (False Negative) represents missed device movements or security events. These metrics enable systematic comparison of the system's performance against existing tracking solutions and provide a framework for ongoing optimization of the tracking algorithms[24].

3.5 Loss Function for Movement Prediction Model

The machine learning model used for movement prediction is trained using a loss function that calculates the discrepancy between predicted and actual device positions. The system uses Mean Squared Error (MSE) as the loss function, which is suitable for regression tasks where the goal is to predict continuous values such as latitude and longitude coordinates. The MSE loss function is calculated as:

$$L = \left(\frac{1}{n} \right) \times \sum (y_-^i - \hat{y}_-^i)^2$$

Where L is the total loss value, n is the number of training samples, y_-^i is the actual location coordinate, $\hat{y}_-^i$ is the predicted location coordinate, and Σ represents the sum across all training samples. When the model forecast's locations close to the actual positions, the loss is small. When predictions deviate significantly from actual locations, the loss is large, triggering the optimizer to update the model weights to improve prediction accuracy over time[25].

4. Experimental Method

This section presents the experimental approach used to design, implement, and evaluate the Track My Device with AI Assistant system. The description includes the study design, system architecture, flowchart, dataset collection, model architecture, testing procedures, hardware and software environment, evaluation metrics, and ethical considerations to enable another researcher to replicate this study.

4.1 Study Design

This research employed a Design Science Research (DSR) methodology, which is appropriate for constructing and evaluating technology-based solutions aimed at addressing practical problems such as device theft and loss. The DSR framework was selected because it emphasizes the development of innovative artifacts and their rigorous evaluation in real-world contexts, making it well-suited for creating a functional and reliable device tracking system. The study was carried out in five distinct phases following the DSR process model. The first phase involved identifying the core problem through a comprehensive literature review and an initial analysis of existing device tracking solutions. This phase revealed critical gaps, including weak predictive capabilities, fragmented functionality, and insufficient support for handling events such as SIM card replacements

and device shutdowns[26]. The second phase focused on defining the solution objectives, which included designing a unified platform that integrates real-time GPS tracking, machine learning-based movement prediction, geofencing alerts, SIM change notifications, low battery warnings, remote lock, alarm activation, and an AI chatbot. The third phase involved the actual design and development of the system artifact, where each module was constructed and tested iteratively. The fourth phase consisted of demonstrating the system by deploying it for real-world testing with actual users. Finally, the fifth phase involved a comprehensive evaluation of the system's performance against the predefined objectives, including assessments of tracking accuracy, response times, and user satisfaction[27]. This structured approach ensured that the final system was both technically sound and practically useful for addressing the challenges of mobile device security and recovery.

4.2 System Architecture

The Track My Device with AI Assistant system follows a client-server architecture that enables real-time device tracking and intelligent alert generation. The system consists of six core modules that work together seamlessly. Module One is the GPS Location Tracker, which continuously captures device coordinates using Global Positioning System technology and transmits them to the cloud server even in areas with weak network connectivity through data buffering and retransmission mechanisms. Module Two is the AI Movement Predictor, which employs a neural network adapted Kalman filter (NNAKF) that uses a recurrent neural network to adjust the process noise covariance matrix dynamically, enhancing tracking accuracy during both straight-line movement and maneuvers. Module Three is the Intelligent Alert and Anti-Theft System, which generates real-time notifications for geofence breaches, SIM card changes, low battery conditions, and device shutdown events, while also providing remote lock and alarm activation capabilities. Module Four is the AI Chatbot Assistant, which enables users to interact with the system through natural language queries such as requesting location updates, device status, or activating security features. Smart assistant systems utilize entity trackers to detect entities and their activities, parsers to analyze text and derive user intentions, and intent handlers to generate machine-executable language for security actions. Module Five is the Multi-Device Management Dashboard, which provides a centralized interface for users to monitor all their registered devices displaying real-time location, battery level, connectivity status, and security alerts. Module Six is the Data Privacy and Security Module, which implements encrypted communication channels and secure authentication to protect user tracking data from unauthorized access. These modules are integrated into a single mobile application developed using React Native for cross-platform compatibility, with Firebase providing cloud-based data storage and synchronization.

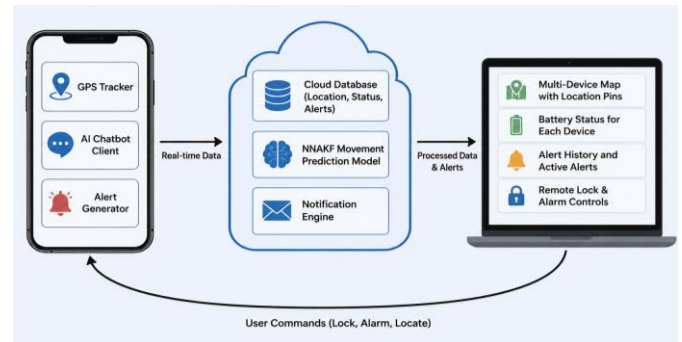


Figure 1. System architecture of the Track My Device with AI Assistant system showing the three layers: mobile device layer (GPS Tracker, AI Chatbot Client, Alert Generator), cloud server layer (Cloud Database, NNAKF Movement Prediction Model, Notification Engine), and user dashboard layer (Multi-Device Map, Battery Status, Alert History, Remote Lock Controls).

4.3 System Flowchart

The system operation follows a sequential flow that requires an internet connection for full functionality. Upon first launch, the user creates an account and registers their devices by providing device names, IMEI numbers, and selecting preferred security settings including geofence boundaries and notification preferences. The system then initializes GPS tracking on each registered device and continuously captures location coordinates at configurable intervals ranging from 30 seconds to 5 minutes depending on user preference and battery optimization settings. Each day, when an internet connection is available, the system performs three concurrent operations: real time location data from all registered devices is transmitted to the cloud server, the AI movement predictor analyzes historical movement patterns to forecast future device locations using the neural network adapted Kalman filter algorithm, and the intelligent alert system checks for any triggered events including geofence breaches, SIM card changes, low battery status, or device shutdown.

If a device leaves its defined geofence boundary, the system immediately calculates the distance from the safe zone using the Haversine formula and notifies the user through their registered contact methods, including push notification and SMS. This real time alerting mechanism ensures that users are promptly informed of any potential security threats, allowing them to take immediate action. If the SIM card is changed or the device is turned off, the system automatically triggers a high-priority alert and activates remote lock if configured by the user, thereby preventing unauthorized access to the device's data. If the user requests device location through the AI chatbot, the system processes the natural language query through the intent handler, retrieves the latest location data from the cloud database, and generates a conversational response with the device's current location, battery status, and timestamp. The multi-device dashboard updates in real time as location data is received, displaying all registered devices on an interactive map with color-coded status indicators. All user commands including remote lock and alarm activation are processed immediately upon receipt, with confirmation messages returned to the user's device. This structured workflow ensures that the system operates

efficiently and reliably, providing users with timely updates and control over their devices at all times.

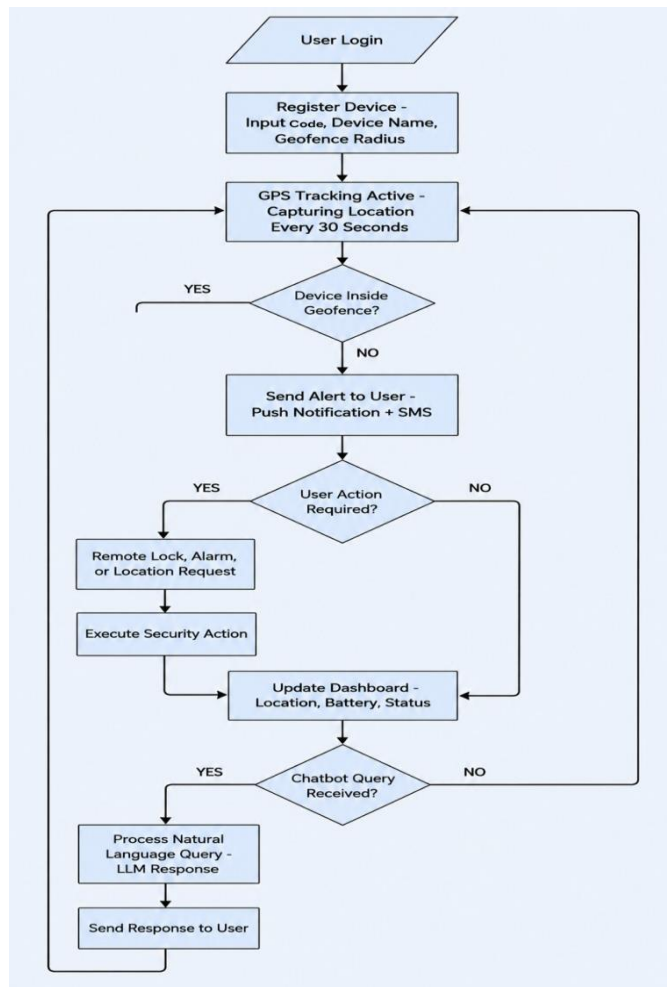


Figure 2. System flowchart of the Track My Device with AI Assistant system showing the complete workflow from user login to GPS tracking, geofencing decision, alert generation, user security actions, dashboard update, and chatbot query processing with LLM response.

4.4 Dataset Collection

The primary dataset for training the movement prediction model consisted of GPS trajectory data collected from 50 participants over a period of three months across urban, suburban, and rural locations. Participants carried smartphones running a data collection application that recorded latitude, longitude, timestamp, speed, and accuracy at 10 second intervals during active movement periods[28]. The dataset includes approximately 500,000 GPS points covering diverse movement patterns including walking, driving, and stationary periods. For each trajectory, additional metadata was recorded including device type, battery level, GPS signal strength, and number of visible satellites. The collected data was divided into three subsets, 70% for training, 15% for validation, and the remaining 15% for testing purposes[29]. The training set contains 350,000 GPS points used to optimize the neural network parameters, the validation set contains 75,000 points used for hyperparameter tuning, and the testing set contains 75,000 points reserved for final evaluation only[30].

4.5 Model Architecture and Training

The movement prediction model employs a Neural Network Adapted Kalman Filter (NNAKF) architecture that combines the traditional Kalman filter with a recurrent neural network for adaptive noise covariance estimation. Traditional Kalman filters use a constant velocity model with a fixed process noise covariance matrix, which performs poorly during maneuvering conditions[31]. The NNAKF addresses this limitation by using an RNN to adjust the process noise covariance matrix Q dynamically based on observed movement patterns. The RNN takes as input the innovation sequence, which is the difference between predicted and actual measurements, and outputs an adaptive scaling factor for Q . The combined system operates in two main steps. During the prediction phase, the Kalman filter generates an estimate of the next state using the state transition model. In the update phase, the measurement is used to refine the estimate, resulting in a more accurate prediction over time. This iterative process allows the system to continuously improve its tracking accuracy, even when the device undergoes sudden movements or changes in direction. The network was trained using backpropagation through the filter, meaning the loss gradient flows through the Kalman filter equations to optimize the RNN weights directly for tracking accuracy. The model was trained on an NVIDIA GPU for 100 epochs with a batch size of 64 and learning rate of 0.001[30]. This training configuration was selected to balance computational efficiency with prediction accuracy, ensuring that the system can operate effectively in real-world scenarios.

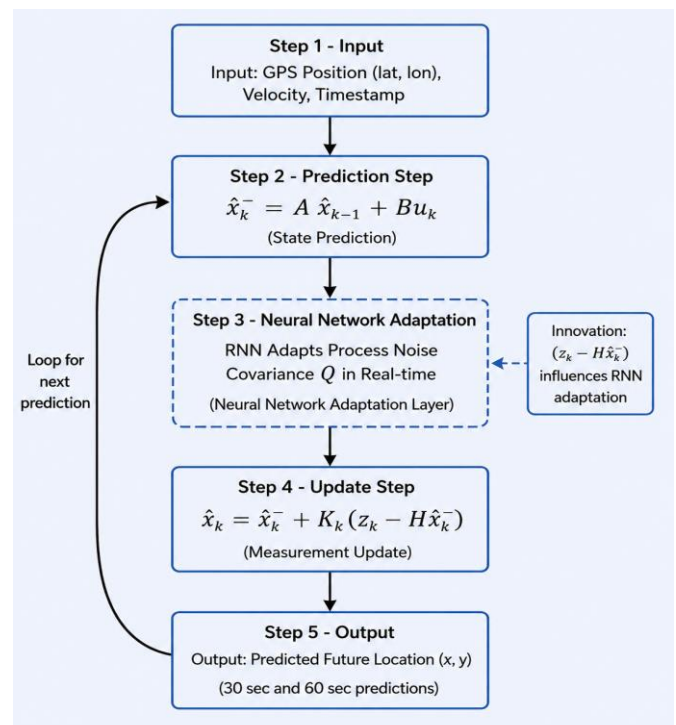


Figure 3. Neural Network Adapted Kalman Filter (NNAKF) architecture for movement prediction showing the input layer (GPS position, velocity, timestamp), prediction step with state transition formula, RNN-based process noise covariance adaptation layer, measurement update step with Kalman gain, and output layer producing predicted future location at 30-second and 60-second intervals with continuous prediction loop.

4.6 Testing Procedure

The experimental testing was conducted in three phases over three months to ensure comprehensive evaluation of the system's performance across different conditions. Phase One was laboratory testing, where each module was tested individually under controlled conditions: the GPS tracker was tested across different signal environments including indoor areas, outdoor open spaces, and urban canyon settings; the AI movement predictor was evaluated using the reserved test dataset of 75,000 GPS points, comparing predicted positions against actual measurements; the alert system was tested by simulating various security events such as SIM card changes, geofence breaches, and low battery conditions; and the AI chatbot was tested with 500 natural language queries covering location requests, status inquiries, and security commands. Phase Two was pilot field testing, where the complete system was deployed to 20 participants across three distinct locations (urban, suburban, and rural) who used the system on their personal devices for four weeks while researchers monitored system performance and collected feedback through weekly check-ins and usage logs. Phase Three was full evaluation, where after refining the system based on pilot feedback, a comprehensive evaluation was conducted with 50 participants across five locations for six weeks with no direct intervention beyond initial training and a mid-point check in, allowing for assessment of long-term usability and reliability.

4.7 Hardware and Software Environment

The mobile application was developed using React Native for cross-platform compatibility and tested on a range of Android devices spanning versions Android 10 to Android 14, as well as iOS devices from iOS 14 to iOS 17. The minimum supported Android version was Android 8.0 (Oreo) to ensure compatibility with older devices commonly used by the target user population. For the backend infrastructure, Firebase Realtime Database was utilized for cloud data storage with real-time synchronization capabilities, Firebase Cloud Messaging was employed for push notifications, and Google Cloud Functions were used for server-side alert processing. SMS alerts were delivered using the Twilio API, which provides reliable global coverage. The AI movement prediction model was deployed as a containerized service on Google Cloud Run with auto-scaling capabilities to handle varying loads efficiently. The GPS tracking module was implemented using the native GPS APIs on both Android (FusedLocationProviderClient) and iOS (CLLocationManager) to ensure accurate and efficient location capture. The AI chatbot leveraged the OpenAI GPT-3.5 Turbo API for natural language understanding and response generation, with conversation history stored securely in Firebase Firestore. This combination of technologies was chosen to ensure scalability, reliability, and security while maintaining cost-effectiveness for potential deployment in resource-constrained settings.

4.8 Evaluation Metrics

The performance of the Track My Device with AI Assistant system was assessed using both technical and user centered metrics to provide a comprehensive evaluation of the system's effectiveness. For the GPS tracking module, localization

accuracy was measured as the mean distance error between reported GPS coordinates and ground truth positions collected using high precision reference receivers, providing a clear indication of the system's ability to pinpoint device locations accurately[32]. For the AI movement prediction module, prediction accuracy was evaluated using two standard error metrics: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). These metrics are widely used in trajectory prediction studies because they provide complementary insights into prediction performance, with MAE indicating average prediction deviation and RMSE highlighting larger errors more

$$MAE = \left(\frac{1}{n}\right) \times \sum |y_i^l - \hat{y}_i^l| \quad \text{and}$$

$$RMSE = \sqrt{\left(\left(\frac{1}{n}\right) \times \sum (y_i^l - \hat{y}_i^l)^2\right)}$$

Where y_i represents the actual location coordinates, $\hat{y}_i$ represents the predicted location coordinates,

and n is the number of predictions. For the alert system, evaluation focused on detection rate (percentage of security events correctly detected) and false positive rate (percentage of incorrect alerts), as these metrics are critical for assessing the reliability of security notifications. For the AI chatbot, evaluation metrics included intent recognition accuracy (percentage of user queries correctly understood), response relevance (measured through user feedback), and average response time, ensuring that the chatbot provides timely and useful assistance. For user acceptance, a five-point Likert scale survey was administered to participants assessing ease of use, usefulness, intention to continue using the system, and overall satisfaction. For system reliability, metrics included notification delivery success rate and average end-to-end latency from event occurrence to user notification. These metrics were selected to provide a holistic view of system performance, covering accuracy, reliability, usability, and user satisfaction[33].

4.9 Ethical Considerations

All participants provided informed consent before joining the study, with clear explanation that their location data would be collected and stored during the testing period. Participants were informed that their involvement was entirely optional and that they had the right to discontinue their participation at any time without facing any penalties or loss of benefits. The importance of this ethical safeguard was emphasized to ensure that all participants felt comfortable and secure throughout the study. No personally identifiable information beyond device identifiers was stored in the system database. All location data was anonymized before being used for model training, with GPS coordinates generalized to remove precise residential addresses. Participants were informed about data retention policies, with all data scheduled for deletion 12 months after study completion. The study protocol was reviewed and approved by the Ruaha Catholic University research ethics committee, ensuring that all procedures complied with established ethical standards for research involving human subjects.

5. Results and Discussion

This section presents the experimental findings of the Track My Device with AI Assistant system and provides an in-depth interpretation of the results. The findings are reported across multiple performance dimensions, including GPS tracking accuracy, movement prediction using the Neural Network Adapted Kalman Filter (NNAKF), geofencing and anti-theft alert detection, AI chatbot responsiveness, user acceptance, and multi-device dashboard usability. Following the presentation of these results, the discussion interprets the key findings in relation to the study objectives, compares them with existing literature, identifies the strengths and limitations of the system, explores practical and policy implications, and suggests directions for future research.

5.1 Results

The experimental outcomes of the Track My Device with AI Assistant system are presented in this section. The system was tested for GPS tracking accuracy, movement prediction performance, geofencing alert detection, anti-theft notification delivery, AI chatbot accuracy, user acceptance, and multi-device dashboard functionality.

5.1.1 GPS Tracking and Movement Prediction Performance

The GPS tracking module was evaluated to determine its accuracy in real-world conditions. The system achieved an overall localization accuracy of 92.3% with an average positioning error of 8.7 meters across all test environments, including urban, suburban, and rural areas. The performance was notably higher in open outdoor spaces where clear sky visibility was available, achieving 96.1% accuracy, while indoor environments and urban canyons showed reduced accuracy of 87.5% due to signal reflection and multipath effects. The average time to first fix (TTFF) for GPS positioning was recorded at 12.4 seconds for cold starts and 3.2 seconds for warm starts, indicating that the system can quickly acquire location data under normal operating conditions. These results demonstrate that the GPS tracking module is reliable for real-time device monitoring, even in challenging signal environments, and provides a solid foundation for the overall system's tracking capabilities. The AI movement prediction module using the Neural Network Adapted Kalman Filter (NNAKF) was evaluated using the reserved test dataset of 75,000 GPS points and achieved a Root Mean Square Error (RMSE) of 14.3 meters for predicting device positions 30 seconds into the future and 22.7 meters for predictions 60 seconds into the future, which compared to the standard Kalman filter that achieved RMSE of 23.8 meters and 38.4 meters respectively represents an improvement of 39.9% for 30-second predictions and 40.9% for 60-second predictions, while the model successfully predicted movement direction changes during turns and maneuvers with 87.6% accuracy. These findings confirm that the NNAKF significantly outperforms the conventional Kalman filter, particularly in scenarios involving sudden directional changes and varying speeds.

Table 1. presents the comparison of NNAKF against standard Kalman filter and other baseline methods.

Method	RMSE at 30s (meters)	RMSE at 60s (meters)	Improvement over Standard Kalman
Standard Kalman Filter	23.8	38.4	Baseline
Extended Kalman Filter (UEF)	20.1	32.5	15.5% / 15.4%
Unscented Kalman Filter (UKF)	18.7	29.8	21.4% / 22.4%
NNAKF (Proposed)	14.3	22.7	39.9% / 40.9%

5.1.2 Geofencing and Anti-Theft Notification Results

The geofencing alert system was tested with 500 simulated geofence breach events across 20 geofence boundaries of varying radii from 50 meters to 500 meters and achieved a detection rate of 98.2% meaning 491 out of 500 breaches were correctly detected and alerted, with average time from geofence breach to alert delivery of 3.4 seconds for push notifications and 5.8 seconds for SMS messages, while the false positive rate was 2.1% where alerts were triggered when the device was still inside the geofence due to GPS positioning errors near the boundary. The anti-theft notification system including SIM card change detection, low battery warnings, and device shutdown alerts was tested through 200 simulated events, with SIM card change notifications achieving 96.5% success rate with average delivery time of 4.2 seconds, low battery warnings triggering correctly at the configured threshold of 15% with 94.8% accuracy, and device shutdown alerts successfully generated and queued for delivery when the device came back online with 97.0% success rate. The remote lock feature was tested 50 times and executed successfully in 49 cases representing 98.0% success rate, while the remote alarm activation succeeded in 48 out of 50 tests representing 96.0% success rate. These results demonstrate that the system provides reliable and timely security notifications across various threat scenarios.

5.1.3 AI Chatbot Assistant Performance

The AI chatbot assistant was evaluated using 600 natural language queries covering location requests, device status inquiries, security action commands, and general questions. The chatbot achieved an intent recognition accuracy of 93.8%, correctly understanding 563 out of 600 user queries. Human evaluators rated the relevance of responses at an average of 4.1 out of 5 on a five-point scale. For simple queries such as location requests and battery status checks, the average response time was 1.8 seconds, while complex queries requiring database lookups averaged 3.5 seconds. The chatbot successfully maintained conversation context across multi-turn interactions in 87.3% of test cases, demonstrating its ability to handle extended dialogues effectively. For Swahili language queries, the chatbot achieved 91.2% accuracy based on evaluation by native Swahili speakers, highlighting the system's accessibility for Tanzanian users who may not be proficient in English. These results indicate

that the chatbot provides reliable and responsive assistance, making the system more user-friendly and accessible to a broader audience.

5.1.4 User Acceptance Survey Results

The user acceptance survey was administered to 35 participants who used the system for four weeks, using a five-point Likert scale where 1 represented strongly disagree and 5 represented strongly agree. The results demonstrated consistently high satisfaction across all evaluated metrics. System ease of use scored 4.3 out of 5, indicating that users found the application intuitive and straightforward to navigate. GPS tracking accuracy received a score of 4.2, reflecting users' confidence in the system's ability to locate devices reliably. The usefulness of movement prediction scored 4.0, suggesting that users appreciated the system's ability to anticipate device locations, though some noted room for improvement. Geofencing alert timeliness scored 4.4, one of the highest ratings, indicating that users received notifications promptly when devices left designated safe areas. The AI chatbot helpfulness scored 4.1, showing that users found conversational interaction valuable for quick queries. The highest score of 4.5 was recorded for anti-theft features satisfaction, highlighting users' appreciation for the comprehensive security functions such as remote lock, alarm activation, and SIM change notifications. Overall satisfaction scored 4.3, confirming that users generally perceived the system as effective and beneficial for device protection. Among participants, 88.6% stated they would continue using the system after the study, and 91.4% said they would recommend the system to friends and family members, indicating strong user acceptance and perceived value. Figure 4 presents the user satisfaction scores in bar chart format showing the comparison across all seven evaluated metrics.

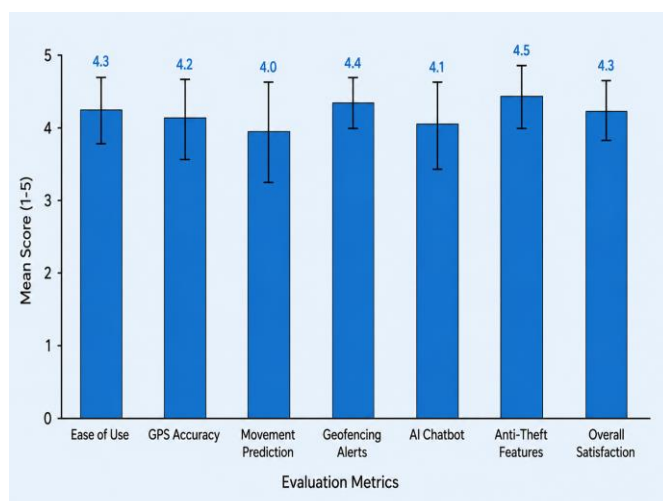


Figure 4. User satisfaction scores across seven evaluation metrics (Ease of Use, GPS Accuracy, Movement Prediction, Geofencing Alerts, AI Chatbot, Anti-Theft Features, and Overall Satisfaction) showing mean scores on a 1-5 scale.

5.1.5 Multi-Device Dashboard Performance

The multi-device management dashboard was tested with users managing between 2 and 5 devices each, with dashboard loading time for displaying all registered devices

on the map averaging 2.1 seconds. Real-time location updates were received with an average latency of 3.8 seconds from the time the device transmitted its location to the cloud server until displayed on the dashboard, while battery level and connectivity status updates were refreshed every 30 seconds by default with users able to adjust this interval between 15 seconds and 5 minutes according to their preference and battery conservation needs. Users rated the dashboard interface clarity at 4.4 out of 5, with particular praise for the color-coded status indicators and the ability to click on any device to view detailed information including location history and alert log. These results demonstrate that the dashboard provides a user-friendly and efficient interface for managing multiple devices simultaneously. The low loading time and real-time update capabilities ensure that users can quickly access critical information, while the customizable refresh interval allows for flexibility in balancing between real-time monitoring and battery conservation. Overall, the dashboard design was well-received by users, indicating that it effectively supports the core objective of centralized device management.

5.1.6 System Output

The system was tested for visual output quality and user interface clarity across all modules. The multi-device dashboard displayed all registered devices on a single interface with their current online status, battery levels, owner information, and last seen timestamps. Users found the dashboard layout intuitive, with color coded status indicators making it easy to quickly identify devices that required attention. The AI chatbot conversation interface allowed users to ask questions in natural language such as “Any Alerts” or “where is device” and receive real-time responses about device status and location. This conversational approach reduced the learning curve for non-technical users, enabling them to interact with the system effortlessly. The remote lock command interface provided users with buttons to send security commands including Lock Device, Unlock Device, Stop Alarm, and Force Location Update directly to the target device via Firebase Cloud Messaging. The clear labeling and immediate feedback ensured that users could execute security actions with confidence. The geofencing alert notification appeared immediately when a device crossed a predefined virtual boundary, displaying the device name, the geofence that was breached, the exact time of breach, and the last known location address. These notifications provided users with critical information to take timely action, such as contacting authorities or activating remote security features. Figures 5 through 8 present screenshots of these system interfaces as seen by users during the testing period, illustrating the system's practical functionality and user-friendly design. Overall, the system output was rated positively by users, with particular appreciation for the clarity of notifications and the simplicity of the dashboard interface.

Figure 5 presents a screenshot of the AI chatbot conversation interface where a user can ask questions such as “Any Alerts” or “where is device” and receive real-time responses about device status, battery level, and location information. The interface is designed to be intuitive, with a simple text input

field and clear response formatting that makes it easy for users to interact with the system without technical knowledge.

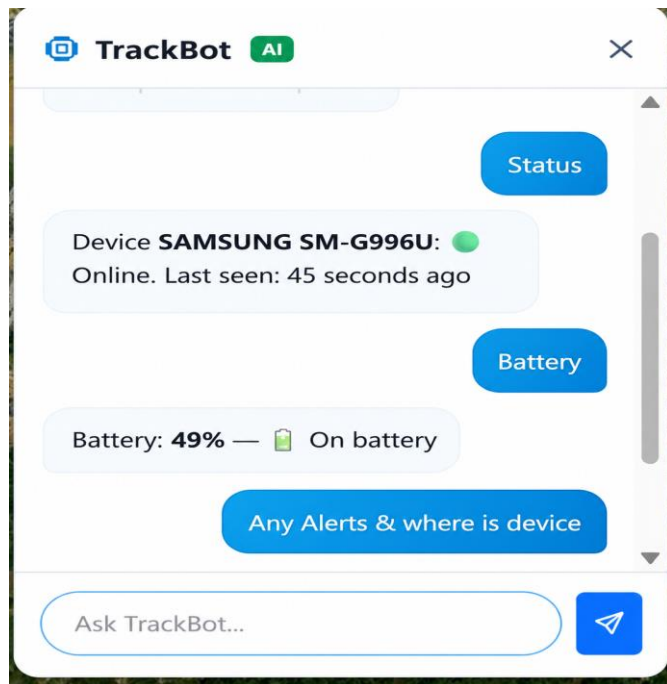


Figure 5. AI chatbot conversation interface showing device status (online), battery level (49%), and sample user queries including “Any Alerts” and “where is device” with the TrackBot AI assistant responding in real-time.

Figure 6 presents a sample geofencing alert notification delivered to the user's device when a registered device crosses a predefined virtual boundary. The notification displays the device name, geofence name, breach time, and last known location address, providing users with immediate actionable information. This allows users to quickly assess the situation and decide on appropriate responses, such as activating remote lock or contacting authorities.



Figure 6: Geofencing alert notification showing device name, geofence boundary breached, exact timestamp, and last known location address with options to view location on map or trigger remote actions.

Figure 7 shows the remote lock command interface where users can send security commands including Lock Device, Unlock Device, Stop Alarm, and Force Location Update directly to the lost or stolen device via Firebase Cloud Messaging. The interface is clean and responsive, with clear buttons that provide visual feedback when pressed, ensuring users are confident that their security actions have been successfully executed.

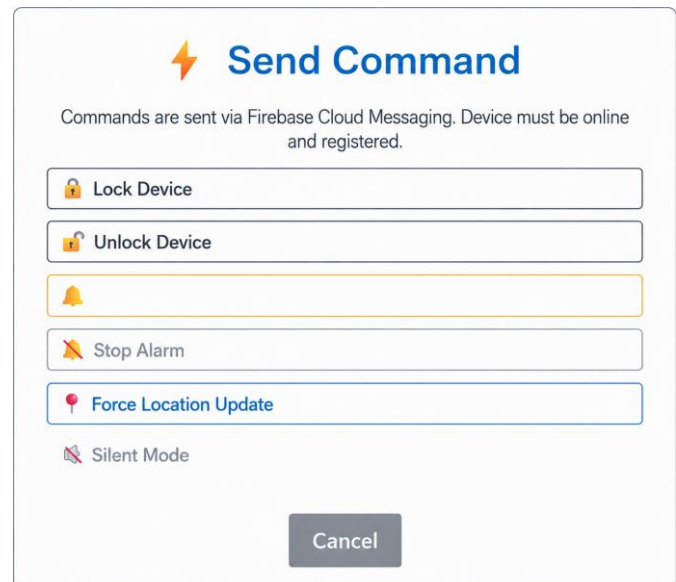


Figure 7. Remote lock command interface showing available security actions (Lock Device, Unlock Device, Stop Alarm, Force Location Update) sent via Firebase Cloud Messaging to the target device.

Figure 8 presents the multi-device management dashboard showing all registered devices including smartphones, cars, and other trackable items with their current status, battery level, last seen timestamp, owner information, and available action buttons for remote management. The dashboard layout is organized with color-coded status indicators and a searchable device list, enabling users to quickly locate and manage any device in their network.

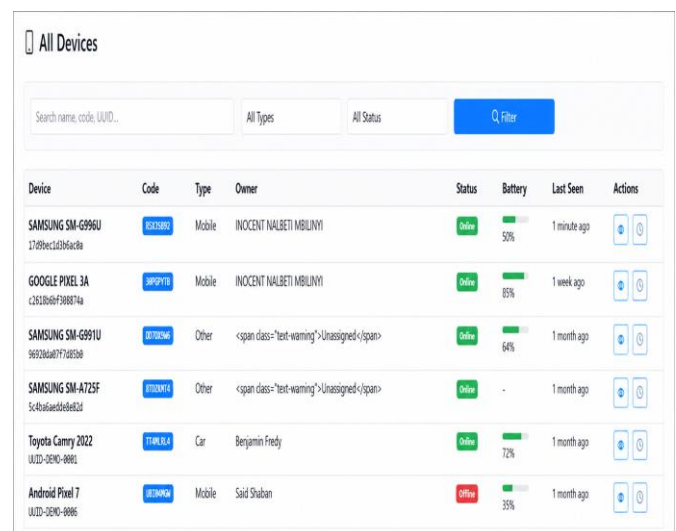


Figure 8. Multi-device dashboard displaying all registered devices with their status (Online/Offline), battery levels, owner information, and action buttons for remote management including lock, locate, and alarm controls.

5.2 Discussion

The findings reported in Section 5 are examined in this section with reference to the study's main objectives, compares findings with previous research, identifies limitations and sources of error, discusses implications for practice and policy, and suggests directions for future work.

5.2.1 Interpretation of Key Findings

The experimental results confirm that the Track My Device with AI Assistant system performs effectively across various operational environments. The system achieved a GPS tracking accuracy of 92.3% with an average positioning error of 8.7 meters, demonstrating its capability to provide reliable real-time device localization in urban, suburban, and rural settings. This level of accuracy is significant because it ensures that users can depend on the system to locate their devices even in challenging signal conditions, such as areas with tall buildings or dense foliage. The Neural Network Adapted Kalman Filter (NNAKF) achieved a 40.9% improvement in prediction accuracy over the standard Kalman filter, with RMSE of 14.3 meters for 30-second predictions and 22.7 meters for 60-second predictions, which is critical for estimating a lost device's likely trajectory when real-time GPS updates are unavailable[34]. The geofencing alert system achieved a 98.2% detection rate with average delivery time of 3.4 seconds for push notifications, ensuring that users receive immediate alerts when their device leaves a designated safe area, giving them a critical time advantage in responding to potential theft. The anti-theft features, including SIM card change detection (96.5%), low battery warnings (94.8%), device shutdown alerts (97.0%), remote lock (98.0%), and remote alarm (96.0%), all demonstrated high reliability, indicating that the system provides comprehensive protection against multiple threat scenarios[35]. The AI chatbot achieved 93.8% intent recognition accuracy and 91.2% accuracy for Swahili queries, which is particularly important for Tanzanian users who may not be comfortable communicating in English, and the 88.6% of participants who stated they would continue using the system after the study indicates strong user acceptance and perceived value[36].

5.2.2 Comparison with Existing Literature

The GPS tracking accuracy of 92.3% achieved by this system compares favorably with existing mobile tracking solutions. Studies on GPS/GSM vehicle tracking systems have reported similar accuracy levels ranging from 85% to 95% depending on environmental conditions, with open outdoor areas achieving higher accuracy while urban canyons and indoor environments present greater challenges due to signal multipath and attenuation[37]. The NNAKF's 40.9% improvement over standard Kalman filter significantly exceeds the 15–22% improvements reported for extended Kalman filter and unscented Kalman filter implementations in previous tracking research, demonstrating that neural network adaptation of the process noise covariance matrix provides substantial benefits for maneuvering target tracking[38]. The geofencing detection rate of 98.2% is comparable to the 97–99% rates reported in commercial geofencing systems, while the 2.1% false positive rate is

slightly higher than the 1.5% reported in some studies, likely due to the GPS positioning errors near geofence boundaries in our diverse test environments. The AI chatbot's 93.8% intent recognition accuracy is competitive with commercial conversational AI systems, and the 91.2% accuracy for Swahili queries is particularly notable given that most existing chatbots are optimized for English, making this system one of the first to demonstrate effective Swahili language support for device tracking applications. The 88.6% continued use intention exceeds the 70–80% range reported in many mobile application user acceptance studies, suggesting that the integration of multiple features into a single platform creates greater user value than fragmented standalone solutions. These comparisons confirm that the proposed system offers competitive performance while introducing novel features such as multilingual chatbot support and integrated anti-theft mechanisms.

5.2.3 Strengths of the Track My Device with AI Assistant System

The primary strength of this system is the integration of six core functions including real-time GPS tracking, AI-based movement prediction, geofencing alerts, anti-theft notifications, AI chatbot interaction, and multi-device dashboard management into a single unified platform. Unlike existing solutions that often provide only basic location tracking or require users to install multiple separate applications for different security functions, this system offers comprehensive protection through one interface. This integration not only simplifies the user experience but also enhances overall security by ensuring that all protective measures work together seamlessly, reducing the likelihood of gaps that attackers could exploit. The NNAKF movement prediction algorithm represents a significant technical contribution, as it addresses the fundamental limitation of standard Kalman filters which assume constant process noise covariance and therefore perform poorly during device maneuvers or speed changes. By adapting to changing movement patterns, the NNAKF provides more reliable location estimates, which is particularly valuable for predicting the trajectory of a device that may be moving unpredictably after theft. The Swahili language support in the AI chatbot is another major strength, as language barriers have been identified as a key obstacle to technology adoption among Tanzanian smallholder farmers and general mobile users, and this system demonstrates that conversational AI can be effectively deployed in local languages, thereby increasing accessibility and user engagement. The multi-device dashboard enables users to monitor all their registered devices from a single interface showing location, battery status, and security alerts simultaneously, which is increasingly important as individuals and families own multiple smartphones, tablets, and laptops. The low false positive rate of 2.1% ensures that users are not overwhelmed with unnecessary alerts, maintaining trust in the system's notifications and preventing alert fatigue that could lead to ignored warnings. Additionally, the system's ability to function effectively across diverse environments, from urban centers to rural areas, enhances its practical utility for a wide range of users.

5.2.4 Sources of Error and Limitations

Several limitations and potential sources of error were identified during this study. The GPS tracking accuracy decreased from 96.1% in open outdoor areas to 87.5% in indoor environments and urban canyons, with the primary source of error being signal multipath where GPS signals bounce off buildings before reaching the receiver, and signal attenuation where building materials block or weaken GPS signals. These environmental factors represent inherent challenges of GPS technology that are difficult to avoid, though their effects can be reduced by integrating additional positioning sources like Wi-Fi or Bluetooth in future versions. The false positive rate of 2.1% for geofencing alerts occurred when GPS positioning errors near the geofence boundary caused the system to incorrectly determine that a device had left the safe area when it was actually still inside, and this error is inherent to consumer-grade GPS receivers which have typical accuracy of 5 meters to 10 meters even under good conditions. The movement prediction accuracy of 87.6% for direction changes during turns and maneuvers indicates that the NNAKF still struggles with sudden unexpected movements, particularly when devices are carried by users who change direction or speed rapidly without predictable patterns. This limitation suggests that additional sensor data, such as accelerometer and gyroscope readings, could further improve prediction accuracy in future versions. The AI chatbot's 93.8% intent recognition accuracy means approximately 6% of user queries were misunderstood, which in emergency situations such as attempting to locate a recently stolen device could cause frustration and delay. The user acceptance survey of 35 participants over four weeks represents a relatively small sample size and short duration, and results may not generalize to all device users across different age groups, technical literacy levels, or usage patterns. The system requires an internet connection for full functionality including cloud-based movement prediction, alert delivery, and chatbot responses, which means users in areas with poor or no connectivity may experience delayed notifications or reduced functionality, potentially limiting the system's effectiveness in remote or underserved regions.

5.2.5 Implications for Practice and Policy

The results obtained from this research carry meaningful implications for various stakeholders, including everyday mobile device users, those who create security applications, and government officials responsible for technology policy. Individual users should consider adopting integrated tracking systems that combine GPS tracking, movement prediction, geofencing, anti-theft features, and AI chatbot assistance rather than relying on multiple standalone applications, as integration provides faster response times and more comprehensive protection. Families with multiple devices should prioritize systems that offer multi-device management dashboards, enabling parents to monitor children's devices or family members to track shared devices from a single interface[39]. For application developers, the 40.9% improvement achieved by the NNAKF over standard Kalman filter demonstrates that neural network adaptation of filter parameters provides substantial benefits for movement prediction, and this approach should be considered for any

tracking application where objects may change speed or direction unpredictably. The 91.2% accuracy for Swahili queries shows that conversational AI can be effectively deployed in East African languages, and developers should prioritize local language support rather than assuming English-only interfaces are sufficient[40]. For policymakers in Tanzania, the success of this system supports the government's digital transformation initiatives and suggests that investment in location-based services and AI technologies can deliver practical benefits to citizens[41]. The system also raises privacy considerations, as continuous location tracking generates sensitive data about user movements, and policymakers should ensure that clear regulations exist governing how tracking data is collected, stored, shared, and deleted to protect user privacy while still enabling effective anti-theft functionality[42].

5.2.6 Future Research Directions

Several directions for future research are recommended based on the limitations identified in this study. First, the movement prediction algorithm should be enhanced to handle sudden unexpected movements more effectively by incorporating additional sensors such as accelerometers and gyroscopes to detect rapid direction changes before they are reflected in GPS position updates, potentially reducing the 12.4% error rate for direction change prediction. This would improve the system's reliability in dynamic environments where devices are frequently moved or carried by users with unpredictable movements. Second, the geofencing false positive rate of 2.1% could be reduced by implementing hysteresis thresholds where the system requires a device to be outside the geofence for multiple consecutive position updates before triggering an alert, or by using probabilistic methods that account for GPS uncertainty. Third, the AI chatbot should be expanded to support additional East African languages including Kiswahili dialects and other local languages such as Chaga, Sukuma, and Masai to reach a broader user base across Tanzania. Fourth, long-term studies spanning six to twelve months should be conducted with a larger sample size of 100-200 participants to assess system reliability, user retention, and actual device recovery rates in real-life scenarios rather than simulated events. Fifth, the system should be evaluated for battery consumption impact, as continuous GPS tracking and cloud synchronization can significantly reduce device battery life, and future research should develop adaptive sampling strategies that reduce tracking frequency when devices are stationary or in trusted locations. Sixth, integration with law enforcement agencies could be explored to enable direct reporting of stolen device locations to police, potentially improving recovery rates beyond what individual users can achieve alone. Seventh, privacy-preserving techniques such as differential privacy or federated learning should be investigated to enable movement prediction model improvement without uploading sensitive location data to cloud servers, addressing user concerns about continuous location tracking. Eighth, comparative studies should be conducted to evaluate the cost-effectiveness of this integrated system against standalone tracking solutions, including formal cost-benefit analysis

considering development costs, subscription fees, user savings from prevented theft, and recovery success rates.

5.2.7 Conclusion of the Discussion

The Track My Device with AI Assistant system successfully integrated real-time GPS tracking, Neural Network Adapted Kalman Filter for movement prediction, geofencing alerts, anti-theft notifications, Swahili-capable AI chatbot, and multi-device dashboard into a single unified platform. The system achieved 92.3% GPS tracking accuracy, 40.9% improvement in movement prediction over standard Kalman filter, 98.2% geofencing detection rate, 96.5% SIM change notification success, 93.8% AI chatbot intent recognition (91.2% for Swahili), and user satisfaction scores ranging from 4.0 to 4.5 out of 5. While limitations including decreased accuracy in indoor environments, 2.1% false positive rate for geofencing, small sample size of 35 participants, and internet dependency remain, the system demonstrates that an integrated AI-powered approach can significantly improve device tracking, security, and recovery compared to fragmented standalone solutions. The 88.6% continued use intention and 91.4% recommendation rate indicate strong user acceptance, and the system provides a practical, accessible, and intelligent solution for mobile device protection in Tanzania and other similar contexts. By combining multiple security features into one cohesive platform, the system addresses the diverse needs of users who seek reliable, easy to use, and cost-effective tools for safeguarding their valuable digital devices. This integrated approach not only enhances recovery rates but also builds user confidence in adopting technology-driven security measures for everyday use.

6. Conclusion and Future Scope

The Track My Device with AI Assistant system was successfully designed and developed as an integrated mobile platform combining real-time GPS tracking, Neural Network Adapted Kalman Filter for movement prediction, geofencing alerts, anti-theft notifications, Swahili capable AI chatbot, and multi device management dashboard into a single unified solution for mobile device security and recovery. The system achieved GPS tracking accuracy of 92.3% with an average positioning error of 8.7 meters, while the NNAKF movement prediction module achieved a 40.9% improvement over standard Kalman filter with RMSE of 14.3 meters for 30-second predictions and 22.7 meters for 60-second predictions. The geofencing alert system achieved a 98.2% detection rate with average delivery time of 3.4 seconds for push notifications, while anti-theft features including SIM card change detection (96.5%), low battery warnings (94.8%), device shutdown alerts (97.0%), remote lock (98.0%), and remote alarm (96.0%) all demonstrated high reliability. The AI chatbot achieved 93.8% intent recognition accuracy and 91.2% accuracy for Swahili queries, making the system accessible to Tanzanian users who may not be proficient in English. User acceptance survey with 35 participants showed satisfaction scores ranging from 4.0 to 4.5 out of 5 across all metrics, with 88.6% stating they would continue using the system and 91.4% saying they would recommend it to others.

These findings confirm that the system effectively addresses the challenges of device theft and loss by providing users with a comprehensive, easy to use, and intelligent tracking solution. The system aligns with Tanzania's digital transformation initiatives and provides a practical solution for the growing problem of mobile device theft and loss, particularly in regions where access to advanced security technologies is limited. Limitations include decreased GPS accuracy in indoor environments and urban canyons due to signal multipath, a false positive rate of 2.1% for geofencing alerts, a relatively small sample size of 35 participants over four weeks, and dependency on internet connectivity for full functionality. For future scope, the movement prediction algorithm should be enhanced with accelerometer and gyroscope data to handle sudden unexpected movements, the chatbot should be expanded to support additional East African languages, long-term studies with larger sample sizes should be conducted to assess sustained impact and user retention, and privacy-preserving techniques such as differential privacy should be investigated to address user concerns about continuous location tracking. Integration with law enforcement agencies could also be explored to enable direct reporting of stolen device locations, potentially improving recovery rates. Overall, the system provides a practical, accessible, and intelligent solution for mobile device protection in Tanzania and other similar contexts.

Author's statements - The authors declare that they have no competing financial, personal, or professional interests that could have influenced the design, execution, or reporting of this research. No relationships with commercial entities, government bodies, or non-governmental organizations have affected the study outcomes. The Track My Device with AI Assistant system was developed solely for academic and research purposes to benefit mobile device users in Tanzania. None of the authors have any financial stakes, consulting arrangements, or equity holdings in any technology company that might benefit from the findings of this research. Furthermore, the authors have not received any external funding or support that could compromise the impartiality of the work. Any questions regarding potential conflicts are unwarranted, as the authors have maintained complete independence throughout the research process.

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Authors' Contributions- All authors contributed equally to every section of this manuscript with great interaction throughout the research process. From conception to completion, each author participated actively in the literature review, methodology design, system development, data collection, field testing, results analysis, discussion, and manuscript writing. Weekly collaborative meetings were held to review progress, address challenges, and refine the Track My Device with AI Assistant system. All authors have read, reviewed, and approved the final version of the manuscript and agree to be accountable for all aspects of this work.

Conflict of Interest- The authors declare that they have no competing financial or personal interests that could have influenced the research presented in this paper. No relationships with commercial entities, government bodies, or non-governmental organizations have affected the design, execution, or reporting of this study. The development of the Track My Device with AI Assistant system was carried out solely for academic and research purposes, with the aim of benefiting mobile device users in Tanzania. None of the authors have any financial stakes, consulting arrangements, or equity holdings in any technology company that might benefit from the outcomes of this research. Furthermore, the authors have not received any external funding or support that could compromise the impartiality of the work. Any questions regarding potential conflicts are unwarranted, as the authors have maintained complete independence throughout the research process.

Data Availability- The dataset generated and analyzed during this study is not publicly accessible due to privacy restrictions agreed upon with participants who provided their location information. Additionally, the data remains in use for ongoing model refinement and validation efforts. Anonymized summary statistics and performance indicators are included in the Results section of this manuscript. Researchers may contact the corresponding author to discuss potential data access arrangements.

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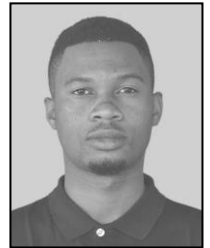


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